

P R O D U C T R E Q U I R E M E N T S
D O C U M E N T

JanSunwai AI

The People's Grievance & Civic Action Platform

Version 1.0
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1. Executive Summary

JanSunwai AI is an end-to-end AI-powered civic resolution engine that transforms how Indian citizens interact with government systems. Named after the traditional concept of public hearings, the platform provides a voice-first, multi-modal interface that enables any citizen — regardless of language, literacy, or internet access — to file grievances, track resolutions, and access their civic rights and entitlements.

The platform serves as a two-sided system: on the citizen side, it provides unprecedented access to grievance redressal, legal rights information, and government scheme discovery; on the government side, it delivers structured civic intelligence, predictive analytics, and accountability metrics that transform urban governance.

Think of it as “Uber for grievances” — citizens can see where their complaint is, who’s handling it, and what’s happening in real-time.

Field	Details
Product Name	JanSunwai AI
Version	1.0
Document Owner	Product Team
Target Users	Indian citizens, Municipal authorities, Smart city administrators
Target Market	Indian Smart Cities (100+ cities under Smart Cities Mission)
Platform	WhatsApp, Voice (IVR/AI), Web Dashboard, Mobile App
Status	Pre-Development / Hackathon Prototype Phase

2. Problem Statement

2.1 The Citizen's Problem

India's civic grievance ecosystem is fundamentally broken. Citizens face a fragmented, opaque, and often hostile system when trying to resolve basic infrastructure and service issues:

- **No single window:** Complaints are scattered across multiple departments, helplines, and portals. Citizens don't know which department handles what.
- **Language and literacy barriers:** Most government portals are English-first and text-heavy, excluding the majority of India's population.
- **Zero accountability:** Complaints disappear into a black hole. There's no tracking, no SLAs, and widespread false closure of tickets.
- **No awareness of rights:** Most citizens have no idea what legal entitlements they have or which government schemes they qualify for.
- **Digital divide:** Internet penetration and smartphone literacy remain barriers for the most vulnerable populations.

2.2 The Government's Problem

City administrators and smart city operators face their own challenges:

- **Unstructured data:** Citizen complaints arrive in unstructured formats across dozens of channels, making pattern detection impossible.
- **Reactive governance:** Without predictive intelligence, cities respond to crises after they escalate rather than preventing them.
- **No citizen feedback loop:** Smart city investments in sensors and dashboards lack the citizen sentiment layer needed to measure actual impact.
- **Resource misallocation:** Budget decisions are made top-down without data on where problems are actually concentrated.

3. Product Vision & Goals

3.1 Vision Statement

"Every citizen, regardless of language, literacy, or internet access, should have an effective voice in governance — and every city should have the intelligence to listen."

3.2 Strategic Goals

Goal	Description	Success Metric
Inclusive Access	Enable any citizen to file and track grievances through voice, WhatsApp, or missed call in their local language	100% of interactions available in 12+ Indian languages; zero-internet filing available
Accountability	Eliminate false closures and create transparent, trackable resolution pipelines	< 5% false closure rate; 90%+ citizen verification of resolved complaints
Civic Empowerment	Surface legal rights, government schemes, and entitlements at the point of grievance	70%+ of interactions include relevant scheme/rights information
Smart City Intelligence	Transform citizen complaints into structured data for predictive urban governance	Real-time data feed into ICCG; predictive alerts with 72hr+ advance warning
Governance Efficiency	Reduce resolution time through AI routing, auto-escalation, and smart utility integration	40% reduction in average resolution time within first 6 months

4. User Personas

4.1 Primary: The Citizen

Attribute	Description
Name	Ramesh / Lakshmi (representative)
Demographics	Age 25–65, urban/semi-urban, Hindi/regional language speaker
Tech Comfort	Low to moderate. Uses WhatsApp and voice calls daily. May not use apps or web portals.
Pain Points	Doesn't know which department to approach; can't track complaints; unaware of legal rights; language barriers
Goals	Get the pothole fixed, get clean water, resolve ration card issues — with minimal effort and maximum transparency
Channels	WhatsApp (primary), Voice calls, Missed call IVR

4.2 Secondary: Municipal Officer / Smart City Admin

Attribute	Description
Name	Priya (Municipal Commissioner) / Arjun (ICCC Operator)
Demographics	Age 30–55, government official, moderate to high tech literacy
Pain Points	Overwhelmed by unstructured complaints; no predictive tools; manual routing; political pressure without data
Goals	Resolve complaints efficiently; identify systemic issues early; justify budget allocation with data; improve ward-level performance
Channels	Web Dashboard, API integration with ICCC

4.3 Tertiary: Proxy Filer / Community Volunteer

Attribute	Description
Name	Sunita (ASHA worker) / Vikram (Local NGO volunteer)
Demographics	Age 25–45, literate, community-embedded, moderate tech skills
Pain Points	Files complaints on behalf of others but has no formal system; no recognition for civic work
Goals	Help community members navigate bureaucracy; earn recognition; see actual results from filed complaints
Channels	WhatsApp, Mobile App, Web Dashboard

5. Feature Architecture

5.1 Module 1: Multi-Modal Intake System

The intake system ensures that no citizen is excluded from filing a grievance due to technology, language, or literacy barriers. It provides five distinct intake channels, each designed for different levels of digital access.

5.1.1 Voice Calls (Inbound & Outbound)

- Citizens call a toll-free or local number and interact with an AI voice agent in their local language.
- AI agent conversationally extracts: issue category, location, severity, and personal details.
- Outbound calls are used for: follow-up on pending complaints, status updates, resolution verification, and scheme notifications.
- Fallback to human agent if AI confidence is below threshold.
- All calls are recorded and transcribed for audit trail.

5.1.2 WhatsApp Bot

- Supports: text messages, voice notes, images, videos, location pins, and documents.
- Conversational flow extracts structured grievance data from unstructured messages.
- Voice notes are transcribed using ASR (Automatic Speech Recognition) in 12+ Indian languages.
- Location pins auto-populate ward and jurisdiction information.
- Rich media messages (images/videos) are processed by the Image Intelligence module.
- Citizens receive a unique complaint ID and can check status anytime by messaging the bot.

5.1.3 Image Intelligence

- Citizens send a photo of the issue (pothole, broken pipe, garbage dump, fallen tree, etc.).
- AI vision model auto-detects: issue category (road damage, sanitation, water, electricity, etc.), estimated severity (minor/moderate/critical), and area type (residential, commercial, highway).
- EXIF data and/or location pin used for geo-tagging.
- Image is stored as evidence and attached to the grievance ticket.
- Before/after photo comparison during resolution verification.

5.1.4 Missed Call Filing (Zero-Internet Access)

- Citizen gives a missed call to a designated number.
- System calls back with an Interactive Voice Response (IVR) menu in their detected/selected language.
- IVR guides through: language selection, issue category, location (by ward name or landmark), and description (voice recording).
- Grievance is filed with zero data/internet consumption on the citizen's end.
- SMS confirmation sent with complaint ID.

5.1.5 Web & Mobile App

- Form-based filing for digitally literate users.
- Map-based location picker.
- Photo/video upload with drag-and-drop.
- Real-time complaint tracking dashboard.
- Push notifications for status updates.

5.2 Module 2: AI Grievance Intelligence Engine

The intelligence layer transforms raw, unstructured citizen input into structured, actionable, and prioritized civic data.

5.2.1 Auto-Classification & Smart Routing

- AI classifies complaints into predefined categories: Water Supply, Electricity, Roads & Potholes, Sanitation & Garbage, Drainage & Sewage, Street Lighting, Public Transport, Ration Card / PDS, Pension & Welfare, Corruption & Misconduct, Building & Construction, Parks & Public Spaces.
- Classification uses a combination of: NLP on text/transcribed voice, image classification from photos, and location-based inference (e.g., complaint near a water treatment plant).
- A knowledge graph of municipal government structure maps each category to the correct department, officer, and jurisdiction.
- Routing is automatic — citizens never need to know which department handles what.

5.2.2 Duplicate & Pattern Detection

- Incoming complaints are matched against existing open complaints using: geo-proximity (within 500m radius), category similarity, time window (within last 7 days), and semantic similarity of descriptions.
- If a match is found, the new complaint is linked to the existing one, and the Community Issue is updated with an increased affected-citizen count.
- Community Issues with high citizen counts get amplified priority scores.

- Pattern detection runs continuously, identifying emerging clusters before they hit critical mass.

5.2.3 Severity Scoring Algorithm

Each complaint receives a composite severity score (0–100) based on weighted factors:

Factor	Weight	Description
Issue Type Base Score	30%	Pre-assigned score per category (flooding = 90, streetlight = 30)
Affected Population	25%	Number of linked complaints / endorsements in the area
Vulnerability Index	20%	Complainant demographics: elderly, disabled, pregnant, BPL household
Time Sensitivity	15%	Seasonal urgency (drainage before monsoon), health risk, safety hazard
Recurrence	10%	How many times same issue reported at same location in past 12 months

5.2.4 Legal Right Mapping

- For each grievance category, the system surfaces relevant legal entitlements from a curated database.
- Sources include: Right to Public Services Act (state-specific), Municipal Corporation Acts, Consumer Protection Act, RTI Act, and relevant Supreme Court / High Court orders.
- Information is presented in simple language in the citizen's local language.
- Example: "Under the Maharashtra Right to Public Services Act, the PWD must respond to road repair complaints within 7 days. Here is the relevant section."

5.3 Module 3: Community Dashboard & Heatmaps

5.3.1 Live Grievance Heatmap

- Real-time map visualization showing complaint density by area, category, and resolution status.
- Color-coded by category (blue = water, red = roads, green = sanitation, etc.).
- Filter by: time period, category, status (open/in-progress/resolved/escalated), ward, and severity level.
- Click on any cluster to see individual complaints, timeline, and assigned officer.
- Publicly accessible — citizens, journalists, researchers, and activists can view city-wide complaint patterns.

5.3.2 Ward-Level Scorecards

- Performance report card for each municipal ward showing: total complaints received, average resolution time, SLA compliance rate, citizen satisfaction score, top 3 issue categories, and month-over-month trend.
- Comparative ranking of wards within the city.
- Publicly visible to create accountability through transparency.

5.3.3 Trend Alerts

- AI monitors complaint patterns in real-time and generates alerts when anomalies are detected.
- Example triggers: 3x spike in water complaints in a single ward within 24 hours (possible pipeline burst), cluster of electricity complaints during summer peak (possible grid stress), sudden increase in sanitation complaints (possible collection vehicle breakdown).
- Alerts are sent to: city ICCO operators, relevant department heads, and affected citizens (advisory).

5.4 Module 4: Resolution Tracking & Escalation Engine

5.4.1 Auto-Escalation Ladder

A time-bound escalation system ensures no complaint stagnates:

Stage	Timeframe	Action
Level 1	0–48 hours	Assigned to ward-level officer. Citizen notified of assignment.
Level 2	48–96 hours (no acknowledgment)	Escalated to department head. Warning notification sent.
Level 3	96 hours–7 days (no action)	Escalated to Municipal Commissioner / Zonal head.
Level 4	7–14 days (no resolution)	Flagged in public dashboard. RTI auto-draft offered to citizen.
Level 5	14+ days	Marked as “Systemic Failure.” Included in weekly city performance report.

Citizens are notified at each escalation stage via their preferred channel (WhatsApp/call/SMS).

5.4.2 Proof-of-Resolution Verification

- When an authority marks a complaint as “resolved,” the system does NOT auto-close it.
- Instead, the citizen receives a verification call/message: “Your complaint about [issue] at [location] has been marked resolved. Has the issue actually been fixed?”

- If citizen confirms → complaint closed with satisfaction score.
- If citizen denies → complaint automatically reopened and escalated one level.
- Before/after photo comparison can be requested from the citizen.
- This single feature addresses one of the biggest real-world problems in Indian governance: false closure of tickets.

5.4.3 RTI Auto-Drafting

- If a grievance exceeds the legally mandated resolution timeline, the system offers to auto-generate a Right to Information (RTI) application.
- The RTI is pre-filled with: complaint details and timeline, the relevant authority and their legal obligation, specific questions about why the complaint was not resolved, and reference to applicable laws and SLAs.
- Citizen can review, edit, and file with one click (or have it physically mailed via partner services).
- This feature democratizes access to India's most powerful transparency tool.

5.5 Module 5: Voice-First & Inclusive Design

5.5.1 Multi-Language Support

Full voice-to-voice and text support in 12+ Indian languages:

Tier	Languages	Support Level
Tier 1 (Launch)	Hindi, English, Tamil, Telugu, Kannada, Bengali	Full voice + text + IVR
Tier 2 (Month 3)	Marathi, Gujarati, Malayalam, Punjabi	Full voice + text + IVR
Tier 3 (Month 6)	Odia, Assamese, Urdu, and additional dialects	Text + IVR, voice in progress

5.5.2 Accessibility Features

- Screen reader compatible web dashboard (WCAG 2.1 AA compliance).
- High-contrast mode and adjustable font sizes.
- Voice-only navigation — entire complaint lifecycle can be completed without visual interface.
- Senior-citizen mode: simplified flows, larger buttons, slower speech synthesis.
- Sign language support via video upload (future phase).

5.5.3 Proxy Filing System

- Authorized volunteers (ASHA workers, panchayat members, NGO workers) can file complaints on behalf of citizens who cannot do so themselves.
- Consent tracking: verbal consent recorded via voice call, or biometric/OTP verification.
- Proxy filers have a separate dashboard to track all complaints filed on behalf of others.
- Recognition system: volunteers earn points and badges for successful resolutions.

5.6 Module 6: Community Mobilization Layer

5.6.1 “Me Too” Endorsement System

- Any citizen can endorse an existing complaint in their area via WhatsApp (“Reply with YES to support this complaint”) or via the heatmap dashboard.
- Endorsement count is visible on the public dashboard.
- Complaints with high endorsement counts receive amplified severity scores and are flagged as community-priority issues.
- This creates democratic, data-backed prioritization.

5.6.2 Volunteer Verification Network

- Local volunteers can physically verify complaints and resolutions.
- Verification involves: visiting the complaint site, taking geo-tagged and time-stamped photos, and submitting a brief verification report via WhatsApp or app.
- Verified complaints receive a “Community Verified” badge on the dashboard.
- Volunteers earn recognition points, badges, and can be featured on a civic leaderboard.

5.6.3 Civic Knowledge Feed

- AI-curated, personalized feed pushed via WhatsApp or app.
- Content includes: new government schemes relevant to the citizen’s profile (income, location, occupation, category), application deadlines and eligibility checkers, local ward meeting schedules and agendas, municipal budget allocations for their area, and public health and safety advisories.
- Content is delivered in the citizen’s preferred language.
- Transforms the grievance touchpoint into an opportunity and information access point.

5.7 Module 7: Impact Analytics for Authorities

5.7.1 Department Performance Dashboard

- Real-time metrics for each department: total complaints, resolution rate, average resolution time, SLA compliance, citizen satisfaction scores, and top bottlenecks.
- Drill-down by: ward, time period, officer, issue sub-category.
- Comparative benchmarking across departments.
- Exportable reports for municipal council meetings and Smart City Mission reviews.

5.7.2 Predictive Maintenance Suggestions

- AI analyzes historical complaint patterns, seasonal trends, infrastructure age, and weather forecasts to predict likely infrastructure failures.
- Example: “Based on complaint patterns over the past 6 months, Ward 14’s drainage system will likely fail during monsoon. Preventive maintenance recommended by [date].”
- Suggestions include: estimated cost of preventive action vs. reactive repair, recommended timeline, and past failure history for the same infrastructure.

5.7.3 Budget Allocation Insights

- Data-driven budget recommendations based on actual citizen-reported data.
- Example: “Ward 7 had 340 sanitation complaints this quarter — 12x the city average. Current sanitation budget allocation for Ward 7 is only 4% of total. Data suggests reallocation.”
- Helps city CFOs and planners make evidence-based budget decisions.
- Quarterly reports generated automatically.

6. Technical Architecture

6.1 System Architecture Overview

The platform follows a microservices architecture with the following high-level components:

- **Intake Layer:** WhatsApp Business API (via Gupshup), Twilio/Exotel for voice/IVR, REST API for web/mobile app.
- **AI Processing Layer:** Claude API for NLP, classification, conversation, and RTI drafting. Claude Vision for image analysis. Sarvam AI / Deepgram for Indian language ASR and TTS.
- **Intelligence Layer:** Grievance classification engine, duplicate detection, severity scoring, pattern detection, and predictive analytics.
- **Data Layer:** PostgreSQL for structured data, Redis for caching and real-time queues, S3-compatible storage for media files.
- **Presentation Layer:** React web dashboard with Leaflet/Mapbox for heatmaps, mobile-responsive design, WhatsApp as primary citizen interface.

6.2 Tech Stack

Layer	Technology	Justification
Voice / Telephony	Twilio / Exotel / Bland.ai	Exotel is India-optimized; Twilio for global reach; Bland.ai for AI voice agents
WhatsApp Integration	WhatsApp Business API via Gupshup	Gupshup is India's leading WhatsApp Business API provider with local language support
AI / LLM	Claude API (Anthropic)	Best-in-class reasoning for classification, conversation, legal text drafting
Computer Vision	Claude Vision API	Multi-modal understanding of civic infrastructure images
Speech (ASR/TTS)	Sarvam AI / Deepgram	Sarvam specializes in Indian languages; Deepgram for low-latency transcription
Maps & Geospatial	Google Maps API + Leaflet / Mapbox	Google Maps for geocoding; Leaflet/Mapbox for custom heatmap visualization
Frontend	React + Tailwind CSS	Component-based architecture with responsive design
Backend	FastAPI (Python)	High-performance async API framework ideal for AI workloads
Database	PostgreSQL + PostGIS	PostGIS extension for geospatial queries on complaint data

Caching / Queues	Redis + Celery	Redis for real-time data; Celery for async task processing (escalations, notifications)
Media Storage	AWS S3 / MinIO	Scalable object storage for images, videos, voice recordings
Monitoring	Prometheus + Grafana	System health monitoring and alerting

6.3 Data Flow

The end-to-end data flow for a typical grievance follows this sequence:

1. Citizen initiates contact via WhatsApp / Voice Call / Missed Call / Web App.
2. Intake Layer receives the message and routes to AI Processing Layer.
3. AI processes: language detection, speech-to-text (if voice), image analysis (if photo), NLP classification.
4. Intelligence Layer: duplicate check, severity scoring, department routing, legal right lookup.
5. Grievance ticket created in PostgreSQL with all metadata, media links, and geo-coordinates.
6. Citizen receives confirmation with complaint ID, assigned department, and legal rights information.
7. Dashboard updated in real-time: heatmap, ward scorecards, department queues.
8. Escalation engine monitors SLA timers and triggers auto-escalation if deadlines pass.
9. Upon resolution marking, verification call/message sent to citizen for confirmation.

6.4 API Architecture

Endpoint Group	Key Endpoints	Description
Intake	/api/v1/grievance/file, /api/v1/grievance/whatsapp-webhook, /api/v1/grievance/voice-webhook	Receive and process new grievances from all channels
Status	/api/v1/grievance/{id}/status, /api/v1/grievance/{id}/timeline	Track complaint status and full resolution timeline
Intelligence	/api/v1/analytics/heatmap, /api/v1/analytics/trends, /api/v1/analytics/predictions	Dashboard data, trend alerts, and predictive analytics
Admin	/api/v1/admin/department/{id}/queue, /api/v1/admin/escalations, /api/v1/admin/reports	Department management, escalation handling, report generation
Citizen	/api/v1/citizen/rights/{category}, /api/v1/citizen/schemes, /api/v1/citizen/rti/generate	Legal rights, scheme discovery, and RTI generation

7. Data Model

7.1 Core Entities

7.1.1 Grievance

Field	Type	Description
id	UUID	Unique grievance identifier
complaint_number	String	Human-readable complaint number (e.g., JSA-2026-MUM-00001)
category	Enum	Issue category (water, roads, sanitation, etc.)
sub_category	String	Specific sub-category (e.g., pothole, pipeline leak)
description	Text	AI-processed structured description
raw_input	JSON	Original unstructured input (text, transcription, etc.)
location	PostGIS Point	Geo-coordinates of the complaint
ward_id	FK	Municipal ward reference
severity_score	Integer (0-100)	Composite severity score
status	Enum	open / acknowledged / in_progress / resolved / reopened / escalated
channel	Enum	whatsapp / voice / missed_call / web / app
language	String	Language of original complaint
media_urls	Array[String]	Photos, videos, voice recordings
citizen_id	FK	Reference to citizen who filed
assigned_department_id	FK	Auto-routed department
assigned_officer_id	FK	Assigned officer (nullable)
community_issue_id	FK (nullable)	Linked community cluster issue
escalation_level	Integer (1-5)	Current escalation stage
resolution_verified	Boolean	Whether citizen confirmed resolution
satisfaction_score	Integer (1-5, nullable)	Citizen satisfaction rating
created_at	Timestamp	Filing timestamp
updated_at	Timestamp	Last update timestamp
resolved_at	Timestamp (nullable)	Resolution timestamp

7.1.2 Citizen

Field	Type	Description
id	UUID	Unique citizen identifier
phone	String	Primary phone number (used as identifier)
name	String (nullable)	Name (optional, extracted from conversation)
preferred_language	String	Preferred communication language
preferred_channel	Enum	whatsapp / voice / sms
ward_id	FK (nullable)	Home ward if known
vulnerability_flags	Array[Enum]	elderly / disabled / bpl / pregnant
total_complaints	Integer	Count of complaints filed
is_proxy	Boolean	Whether this is a proxy filer
created_at	Timestamp	First interaction timestamp

7.1.3 Community Issue

Field	Type	Description
id	UUID	Unique cluster identifier
title	String	AI-generated summary title
category	Enum	Issue category
centroid	PostGIS Point	Geographic center of the cluster
radius_meters	Integer	Cluster radius
linked_grievance_count	Integer	Number of individual complaints
endorsement_count	Integer	Number of citizen endorsements
severity_score	Integer (0-100)	Amplified severity score
status	Enum	open / in_progress / resolved
created_at	Timestamp	Cluster creation timestamp

8. Non-Functional Requirements

Requirement	Specification	Rationale
Availability	99.9% uptime SLA	Civic infrastructure must be always-on
Response Time (API)	< 500ms for 95th percentile	Real-time dashboard updates
Response Time (AI)	< 3 seconds for classification	Acceptable wait during conversation
Voice Latency	< 1 second for speech processing	Natural conversational experience
Concurrent Users	10,000+ simultaneous connections	City-scale deployment
Data Retention	Minimum 5 years for all grievance data	Legal and audit requirements
Scalability	Horizontal scaling for intake and processing layers	Multi-city deployment readiness
Security	AES-256 encryption at rest, TLS 1.3 in transit	Citizen PII protection
Privacy	DPDP Act 2023 compliance	Indian data protection law compliance
Accessibility	WCAG 2.1 AA compliance	Inclusive design for disabled citizens
Localization	12+ Indian languages at launch	Pan-India accessibility
Backup	Daily automated backups, 30-day retention	Data safety and disaster recovery

9. Unique Differentiators

Differentiator	Why It Matters
Complaint Clustering into Community Issues	Transforms individual frustration into collective civic power. 50 individual complaints become one amplified community issue with 50x the political weight.
Legal Right Auto-Mapping	Empowers citizens with knowledge they never had access to. For the first time, a citizen filing a water complaint learns their exact legal entitlement and timeline.
Auto-RTI Generation	Democratizes India's most powerful transparency tool. Citizens who've never heard of RTI can now file one with a single click.
Missed Call → Callback IVR	True last-mile inclusion. Zero internet, zero literacy, zero smartphone required. Just a phone and a missed call.
Proof-of-Resolution Verification	Addresses the single biggest problem in Indian civic governance: false closure of tickets. No complaint is closed without citizen confirmation.
Predictive Crisis Detection	Turns reactive complaint management into proactive urban governance. Cities can prevent crises instead of responding to them.
Citizens as Smart City Sensors	Every citizen interaction generates structured geospatial, temporal, and categorical data that feeds into smart city infrastructure.

10. Hackathon Demo Strategy

For the hackathon demonstration, the following end-to-end flow will be showcased, covering both Problem Statements:

10.1 Demo Flow

Step	Action	What It Demonstrates	PS Alignment
1	Citizen sends a WhatsApp photo of a pothole with a location pin	Multi-modal intake, image intelligence, geo-tagging, zero-friction filing	PS1: Accessible, inclusive access
2	AI classifies the image, estimates severity, identifies category, and files a grievance	AI classification, auto-routing, structured data extraction	PS1 + PS2: Smart processing
3	Citizen receives complaint ID + legal rights info + relevant scheme notification	Legal right mapping, scheme discovery, information access	PS1: Empowerment & information access
4	Voice call in Hindi: citizen reports a water issue, AI converses and creates ticket	Voice-first, multi-language, conversational AI	PS1: Language inclusion
5	Dashboard shows both complaints on a live heatmap with severity scores	Geospatial visualization, pattern detection, real-time data	PS2: Smart city intelligence
6	System detects cluster: 12 water complaints in Ward 5 in 2 hours	Duplicate detection, community issue creation, trend alert	PS2: Predictive governance
7	Auto-escalation simulation: complaint moves through escalation levels	Accountability engine, SLA enforcement	PS1 + PS2: Governance efficiency
8	Verification callback to citizen asking if the issue was actually fixed	Proof-of-resolution, false closure prevention	PS1: Citizen empowerment

10.2 Demo Duration Target

Total demo time: 5–7 minutes, covering the complete citizen journey from complaint to verified resolution, while demonstrating both citizen-facing (PS1) and smart-city-facing (PS2) capabilities.

11. Success Metrics & KPIs

11.1 Citizen-Side Metrics (PS1)

Metric	Target	Measurement
Grievance filing success rate	> 95%	% of initiated interactions that result in a filed complaint
Language coverage	12+ languages	Number of languages with full voice + text support
Zero-internet filing rate	> 10% of total	% of complaints filed via missed call IVR
Legal rights surfaced	> 70% of interactions	% of grievances where relevant legal entitlements are shown
Scheme discovery	> 30% of interactions	% of interactions where a relevant scheme is surfaced
Citizen satisfaction	> 4.0 / 5.0	Average satisfaction score from resolution verification
False closure rate	< 5%	% of “resolved” complaints rejected by citizens during verification

11.2 Smart City Metrics (PS2)

Metric	Target	Measurement
Average resolution time	40% reduction in 6 months	Median days from filing to verified resolution
Predictive alert accuracy	> 60%	% of predictive alerts that correspond to actual infrastructure issues
Predictive alert lead time	> 72 hours	Average advance warning time before crisis events
Department SLA compliance	> 80%	% of complaints resolved within mandated timelines
Data feed uptime to ICCC	> 99.5%	Availability of real-time data pipeline to city command center
Community issue detection rate	> 90%	% of related complaints correctly clustered
Budget report adoption	Quarterly reports generated	Number of quarterly budget insight reports produced and consumed

12. Risks & Mitigations

Risk	Severity	Mitigation
Government adoption resistance	High	Start with progressive municipalities and smart city SPVs. Demonstrate value through pilot data. Position as a tool that helps officials, not replaces them.
AI misclassification	Medium	Human-in-the-loop review for low-confidence classifications. Continuous model fine-tuning with feedback data. Citizen can correct category.
False/spam complaints	Medium	Phone number verification. Rate limiting. AI-based spam detection. Community verification layer.
Language model accuracy	Medium	Partner with Sarvam AI for Indian language specialization. Continuous evaluation across dialects. Fallback to human agent.
Data privacy concerns	High	DPDP Act compliance. Minimal data collection. Encryption at rest and transit. Citizen consent management. Regular security audits.
Scalability under load	Medium	Horizontal scaling architecture. Load testing before city-wide launch. Queue-based processing for async tasks.
Political misuse	Medium	Data transparency (public dashboards). Tamper-proof audit logs. No manual override of severity scores.
Citizen trust and adoption	High	Start with high-visibility quick wins (potholes, streetlights). WhatsApp-first approach (familiar channel). Word-of-mouth through verified resolutions.

13. Development Roadmap

Phase	Timeline	Deliverables
Phase 0: Hackathon MVP	Week 1–2	WhatsApp bot (text + image), voice call agent (Hindi + English), basic dashboard with heatmap, auto-classification and routing, resolution verification flow
Phase 1: Pilot Launch	Month 1–3	Full 6-language support, missed call IVR, auto-escalation engine, legal rights database, department admin dashboard, pilot with 1 smart city
Phase 2: Scale	Month 3–6	12+ languages, RTI auto-drafting, community issue clustering, predictive analytics, volunteer verification network, ICCC integration API, scheme discovery engine
Phase 3: Platform	Month 6–12	Multi-city deployment, IoT data fusion, digital twin feed, budget allocation insights, open data API for researchers, mobile app (citizen + volunteer)

14. Appendix

14.1 Glossary

Term	Definition
ICCC	Integrated Command and Control Center — the smart city nerve center that aggregates data from multiple urban systems
IVR	Interactive Voice Response — automated phone menu system
ASR	Automatic Speech Recognition — converting speech to text
TTS	Text-to-Speech — converting text to spoken audio
RTI	Right to Information — Indian law enabling citizens to request government information
SLA	Service Level Agreement — mandated response/resolution timelines
DPDP Act	Digital Personal Data Protection Act 2023 — India's data privacy law
PostGIS	PostgreSQL extension for geographic information system (GIS) data
Ward	The smallest administrative division in an Indian city's municipal governance
BPL	Below Poverty Line — economic classification used for government scheme eligibility
ASHA	Accredited Social Health Activist — community health worker in India's public health system
PMKVY	Pradhan Mantri Kaushal Vikas Yojana — government skill development program

14.2 References

- Smart Cities Mission, Ministry of Housing and Urban Affairs, Government of India
- Right to Information Act, 2005
- Digital Personal Data Protection Act, 2023
- Right to Public Services Acts (State-specific legislations)
- WCAG 2.1 Accessibility Guidelines
- WhatsApp Business API Documentation
- Anthropic Claude API Documentation

End of Document